FUNCTIONS

CSE 2213/CSI 219 – Discrete Mathematics

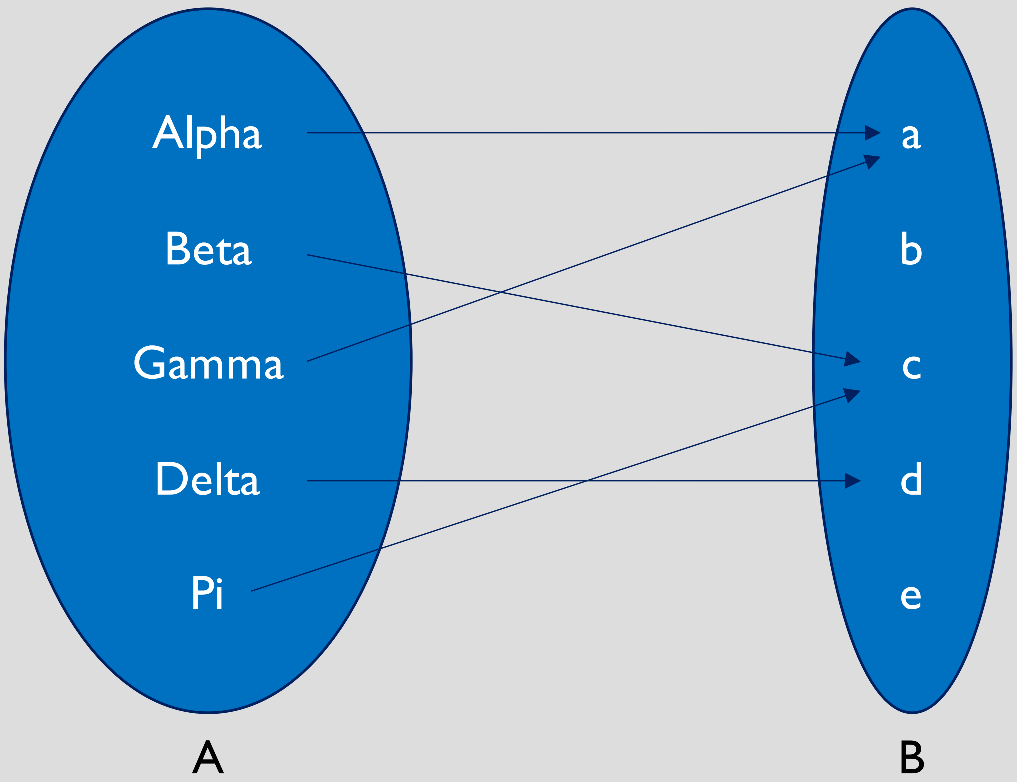
FUNCTION

- Let A and B be nonempty sets. A *function* f from set A to set B is the mapping of every element of A to a single element of B.
- We write $f(a) = b$ if the function named f maps the value of a (a belongs to Set A) to a value of b (b belongs to Set B).
- The above notation is the "Value Notation" of writing functions
- If f is a function from set A to set B, we write

$$f{:}A \rightarrow B.$$

- The above notation is the "Set Notation" of writing functions.

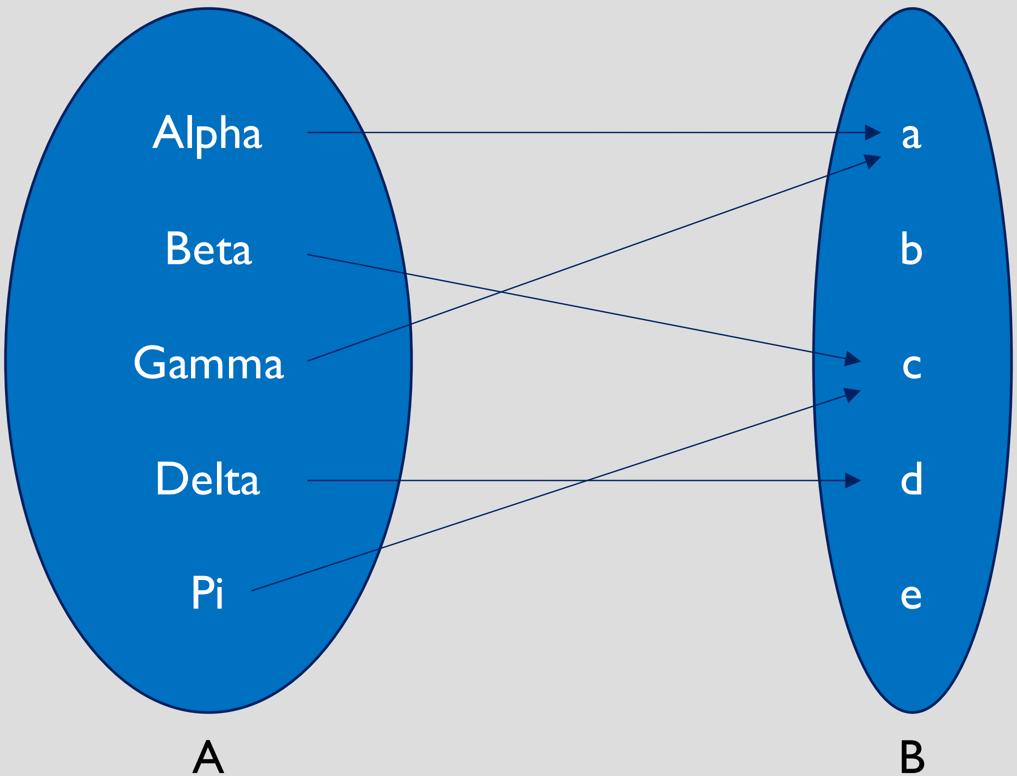
FUNCTION



$f: A \to B$

$f(Alpha) = a$
$f(Beta) = c$
$f(Gamma) = a$
$f(Delta) = d$
$f(Pi) = c$

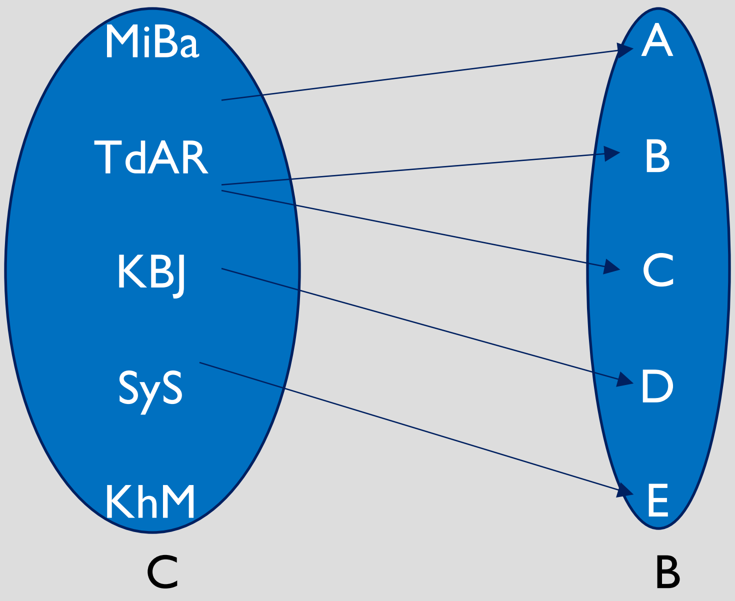
FUNCTION



$f: A \to B$

Domain = Set A
Codomain = Set B
Range/Image = $\{a, c, d\}$

NOT A FUNCTION



$$f: C \rightarrow B$$

- Mapping of $x \in C$ must be unique
  - Here, TdAR is mapped to two different values

- All $x \in C$ must be mapped
  - Here, the value KhM in the domain is not mapped to any value in the codomain

MATHEMATICAL EXAMPLE

- Function: $f: R \rightarrow R, f(x) = x^2$
  - R represents the set of Real Numbers. The "Set Notation" is telling us that the domain consists of Real Numbers, and the codomain consists of Real Numbers too. The Value Notation of the function is telling us that for a input value of x into the function, it outputs/maps a value of x^2
- Function: $f: R \rightarrow R, f(x) = x^2 + 1$

ADDITION AND MULTIPLICATION OF FUNCTION

- Functions of same domain can be added or multiplied
- $(f_1 + f_2)(x) = f_1(x) + f_2(x)$
- $(f_1 f_2)(x) = f_1(x) f_2(x)$
- If $f(x) = x^2$ and $g(x) = x - x^2$,
  then find out $(f + g)(x)$ and $(fg)(x)$

ONE-TO-ONE FUNCTION OR INJECTION

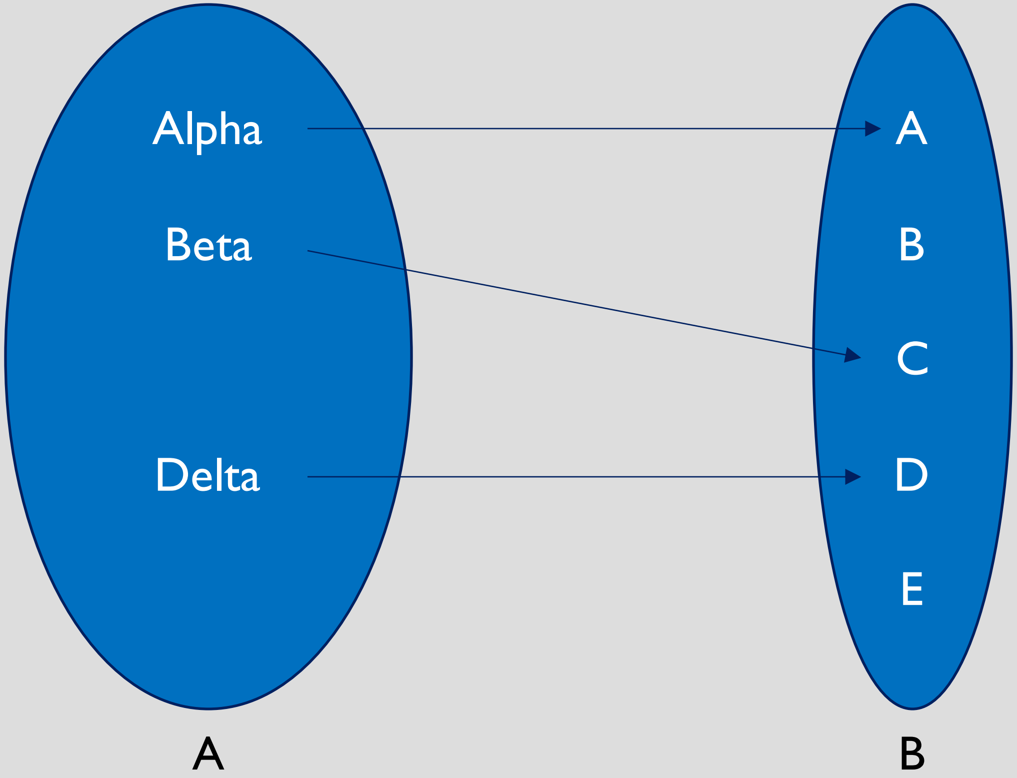
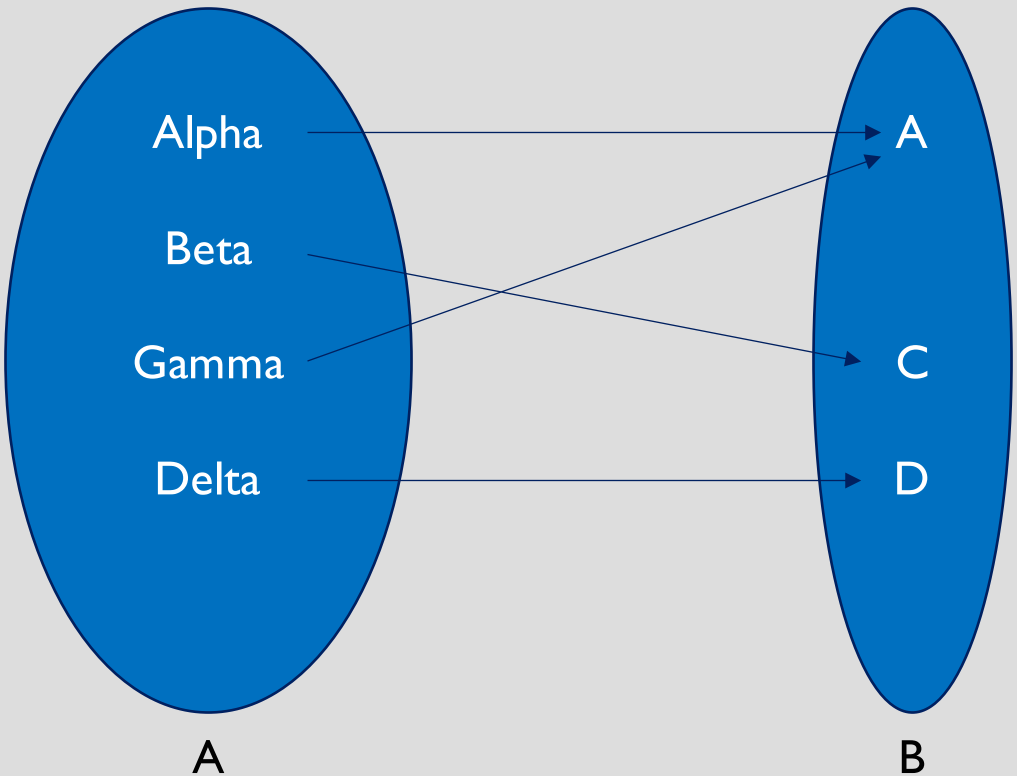


Image of every $x \in A$ must be unique

The value of every image (that is, every member of the **range**) corresponds to only one value of the domain

Note that B and E are images of none

ONTO FUNCTION OR SURJECTION

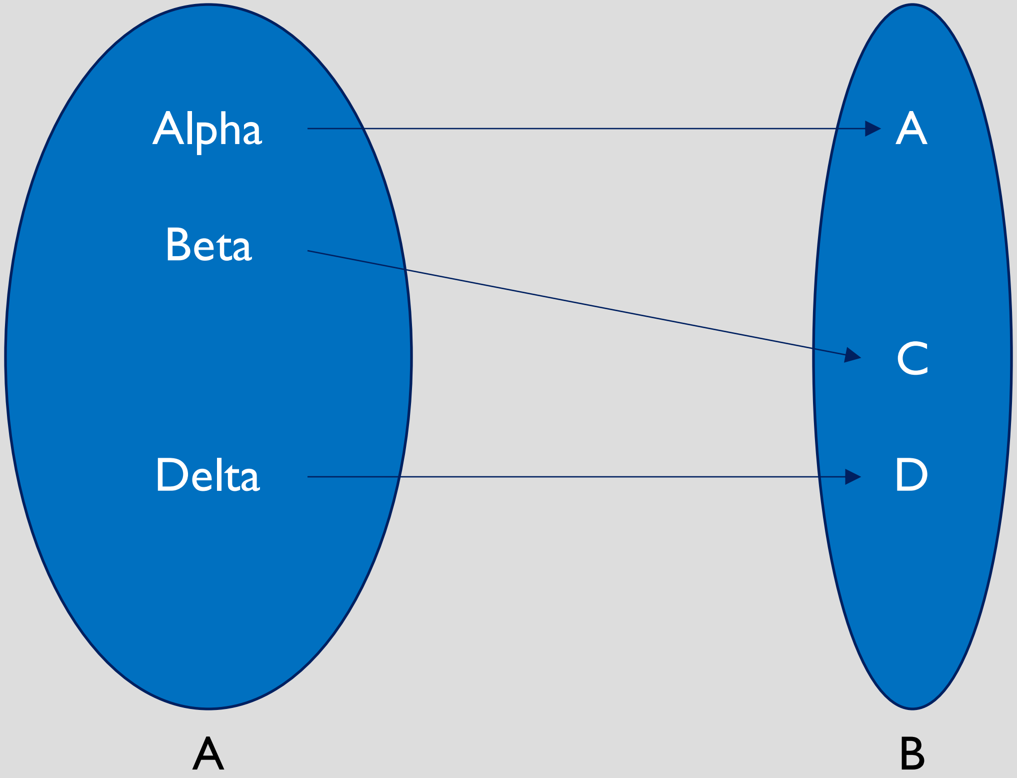


Every $y \in B$ must be an image of some $x \in A$, that is codomain = range

Every value of the codomain can be mapped from some value in the domain.

Note that Alpha and gamma have the same image. This violates the one-to-one property. This is therefore, not a one-to-one function.

BIJECTION



Function that is both one-to-one and onto

PROBLEM

- Find out if the function $f: Z \to Z, f(x) = x^2$ is one-to-one and/or onto.
- Find out if the function $f: R \to R, f(x) = x + 1$ is one-to-one and/or onto.

ONE-TO-ONE AND ONTO FUNCTIONS

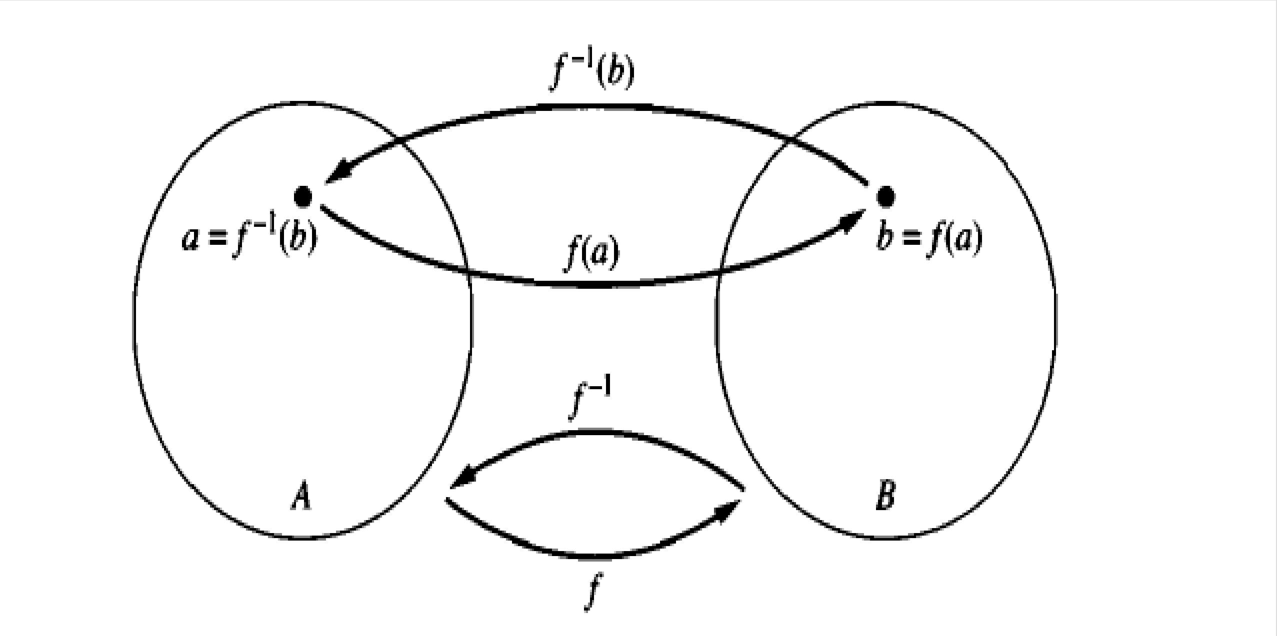
- A function can change its properties with the change of domains
- $f\colon R^+ \to R, f(x) = x^2$ is one-to-one but not onto.
  - Every possible value of the image/range comes from exactly one value in the domain. Therefore, one-to-one.
  - The codomain consists of all real numbers, and the negative numbers of the codomain are never mapped from the domain. Therefore, not onto, since all members of codomain are not covered.
- $f\colon R \to R^+, f(x) = x^2$ is onto but not one-to-one
  - Every possible value of the image can come from **two** numbers in the domain – positive and negative. Therefore, not one-to-one
  - The codomain consists of positive real numbers only, and every value of the codomain can be mapped from the domain. Therefore, onto
- Quiz: Is $f\colon R^+ \to R^+, f(x) = x^2$ a bijection?

INVERSE OF FUNCTIONS

- Inverse of a function exists if the function is a bijection
  - Called an invertible function

Let f be a one-to-one correspondence from the set A to the set B. The *inverse function* of f is the function that assigns to an element b belonging to B the unique element a in A such that $f(a) = b$. The inverse function of f is denoted by f^{-1}. Hence, $f^{-1}(b) = a$ when $f(a) = b$.

EXAMPLE



EXAMPLE

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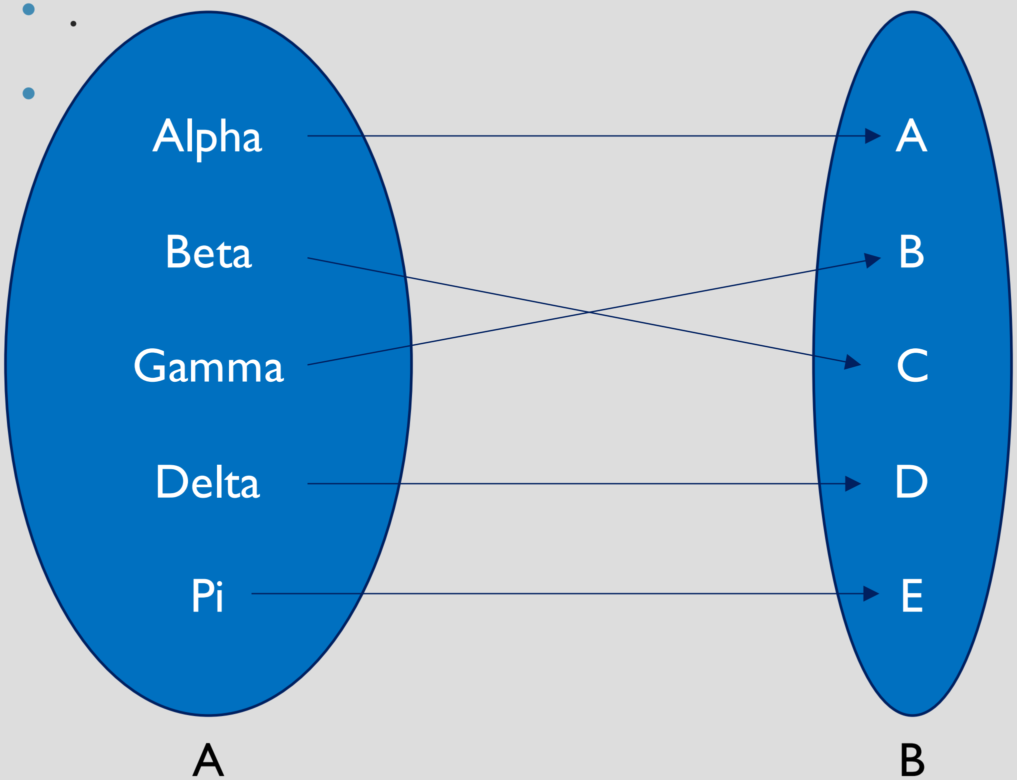

f:A $\rightarrow$ B

$$f(Alpha) = A$$
$$f(Beta) = C$$
$$f(Gamma) = A$$
$$f(Delta) = D$$
$$f(Pi) = C$$

f^{-1}(A)= ???
f^{-1}(C)=
f^{-1}(D)=
f^{-1}(B)= ???

EXAMPLE

Let f be the function from $\{a, b, c\}$ to $\{1, 2, 3\}$ such that $f(a) = 2$, $f(b) = 3$, and $f(c) = 1$. Is f invertible, and if it is, what is its inverse?

Solution: The function f is invertible because it is a one-to-one correspondence. The inverse function f^{-1} reverses the correspondence given by f, so $f^{-1}(1) = c$, $f^{-1}(2) = a$, and $f^{-1}(3) = b$. ◀

EXAMPLE

Let f be the function from $\mathbf{R}$ to $\mathbf{R}$ with $f(x) = x^2$. Is f invertible?

THE COMPOSITION OF A FUNCTION

Let g be a function from the set A to the set B and let f be a function from the set B to the set C. The *composition* of the functions f and g, denoted by $f \circ g$, is defined by

$$(f \circ g)(a) = f(g(a)).$$

EXAMPLE

- Let f and g be the functions from the set of integers to the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. What is the composition of f and g? What is the composition of g and f?
- $f \circ g(x)=f(g(x))=f(3x+2)=2(3x+2)+3=6x+7$
- $g \circ f(x)=g(f(x))=g(2x+3)=3(2x+3)+2=6x+11$

EXAMPLE

- Let g be the function from the set $\{a, b, c\}$ to itself such that $g(a) = b$, $g(b) = c$, and $g(c) = a$. Let f be the function from the set $\{a, b, c\}$ to the set $\{1, 2, 3\}$ such that $f(a) = 3$, $f(b) = 2$, and $f(c) = 1$. What is the composition of f and g, and what is the composition of g and f?